

Course Code:CS2406		
Name of the Programme: B.Tech (Hons.) CSE		
Name of the Course: Incident Response and Digital Forensics		
Semester: IV		
Course Credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
3	1+1+2	60
Course Objectives		
A. Understand the principles of incident response, forensic fundamentals, and legal considerations involved in digital investigations.		
B. Learn methods for collecting and preserving network and host-based evidence, including volatile and non-volatile data.		
C. Explore forensic imaging techniques and gain skills in analyzing network, memory, and storage-based digital evidence.		
D. Develop the ability to document forensic findings through formal reports and apply threat intelligence for proactive cyber defense.		

Syllabus

Module - 1: Understanding Incident Response	No. of hours
	3
The Incident Response Process, The Incident Response Framework, The Incident Response Plan, Incident Response Playbooks, Testing the Incident Response Framework, Engaging the Incident Response Team, Incorporating crisis communications.	
Module - 2: Fundamentals of Digital Forensics and Investigation	No. of hours
Methodology	3
Forensic science overview, Trace evidence and digital forensics, Legal issues in digital forensics, Forensic procedures in incident response, AI in digital forensics. Incident investigation analysis types, Functional Digital Forensic Investigation methodology, Cyber Kill Chain, Diamond Model of Intrusion analysis	
Module - 3: Collecting Network Evidence and Acquiring Host-Based Evidence	No. of hours
	3
Network evidence overview, Firewalls and proxy logs, NetFlow, Packet capture Wireshark, Evidence collection. Preparation, Order of volatility, Evidence acquisition, Acquiring volatile memory, Acquiring non-volatile evidence	
Module - 4: Analyzing System Memory and Analyzing System Storage	No. of hours
	3
Analyzing System Memory, Memory evidence overview, Memory analysis. Analyzing System Storage, Forensic platforms	

Module - 5: Forensic Reporting and Threat Intelligence	No. of hours
	3
Forensic Reporting, Documentation overview, Incident tracking, Written reports, Threat Intelligence, Threat intelligence overview, Threat intelligence methodology, Threat intelligence direction, Threat intelligence sources, Threat intelligence platforms	

Course Outcomes: After completing the course, the students will be able to:	
a.	Demonstrate the core concepts of incident response, digital forensics, and related legal considerations.
b.	Apply standard techniques for collecting and preserving digital evidence from networks and host systems.
c.	Analyze forensic images, memory dumps, and network data to extract relevant information.
d.	Prepare structured forensic reports and incorporate threat intelligence into incident response.

Text Books	
1	G. Johansen, <i>Digital Forensics and Incident Response: A Practical Guide to Deploying Digital Forensic Techniques in Response to Cyber Security Incidents</i> , 1st ed. Birmingham, U.K.: Packt Publishing, 2017. ISBN: 1787288684.
2	G. Johansen, <i>Digital Forensics and Incident Response</i> , 4th ed. Birmingham, U.K.: Packt Publishing, 2025. ISBN: 9781836200109.

Reference Books	
1	G. Johansen, <i>Digital Forensics and Incident Response</i> , 2nd ed. Birmingham, U.K.: Packt Publishing, 2020. ISBN: 183864900X.
2	R. Botwright, <i>Cyber Incident Response: Counter Intelligence and Forensics for Security Investigators</i> . [Place of publication not specified]: Pastor Publishing Ltd, 2024. ISBN: 183938803X.

Tutorials	[15 hours]
1	Simulating a Phishing Incident and Incident Response Scenario: Detect and respond to a phishing attack targeting employee emails. Learn how to identify phishing, contain the threat, and communicate with stakeholders.
2	Creating an Incident Response Plan for a Small Business Scenario: Draft a complete IR plan tailored for a company with 50 employees. Focus on roles, escalation paths, and crisis communication.
3	Building Playbooks for Malware and Ransomware Attacks Scenario: Develop stepwise playbooks for isolating infected machines and recovering data.
4	Conducting Tabletop Exercises for Incident Response Testing Scenario: Run a mock ransomware attack drill with a cross-functional team.
5	Using Ticketing Systems for Incident Tracking and Documentation

	Scenario: Log, update, and close incident tickets in a simulated SOC environment.
6	Writing a Forensic Report After a Data Breach Scenario: Document evidence from a simulated breach, creating timelines and impact summaries.
7	Log Analysis for Incident Timeline Reconstruction Scenario: Analyze system and network logs to recreate the sequence of an intrusion.
8	Collecting Threat Intelligence from Open Sources Scenario: Use AlienVault OTX and VirusTotal to gather intelligence on an emerging malware campaign.
9	Mapping Threat Data to MITRE ATT&CK Framework Scenario: Analyze attacker behaviors using collected data and classify them per ATT&CK tactics.
10	Creating Threat Intelligence Reports for SOC Analysts Scenario: Compile threat data into actionable intelligence reports for security teams.
11	Integrating Threat Intelligence Feeds into SIEM Alerts Scenario: Configure SIEM to automatically generate alerts from threat feeds.
12	Threat Hunting Using IOC Data Scenario: Use Indicators of Compromise (IOCs) to search network data for hidden threats.

Lab Programs		[30 Hours]
1	Incident Response Simulation Using TheHive: Simulate a malware incident and manage the response lifecycle.	
2	Draft an Incident Response Plan Document: Write an IR plan document with templates and roles for a fictional company.	
3	Develop an Incident Response Playbook for DDoS Attacks: Create detailed playbook steps to mitigate and respond to a DDoS attack.	
4	Tabletop Exercise for Social Engineering Attack Response: Conduct a role-playing exercise responding to a social engineering scenario.	
5	Incident Ticket Management in Jira or RT: Create, track, and close tickets related to simulated security events.	
6	Write a Forensic Report Based on Simulated Attack Artifacts: Analyze provided logs, extract evidence, and write a formal forensic report.	
7	Reconstruct an Attack Timeline Using ELK Stack: Ingest logs into ELK and visualize the sequence of an intrusion.	
8	Collect and Analyze Threat Intelligence from OTX and VirusTotal: Extract relevant threat data and analyze malicious IPs/domains.	
9	Use MITRE ATT&CK Navigator to Classify Threats: Map collected indicators to specific ATT&CK tactics and techniques.	
10	Prepare a Threat Intelligence Report for SOC Operations: Use real threat data to prepare and present a threat report.	

11	Configure Open Source SIEM (Security Onion) to Use Threat Feeds: Integrate a threat intelligence feed and validate generated alerts.
12	Threat Hunt with Zeek (Bro) IDS Using IOC Indicators: Search network traffic captures for known malicious indicators.